

1 Process and Thread (3 Questions)

```
int ret = fork();
if(ret == 0) {
    exec(NEW_COMMAND); % Note that NEW_COMMAND does not print out anything
    printf("Child\n");
}else {
    int re = wait(NULL);
    printf("Parent\n")
}
```

What is the output of the above snippet? Assume no error.

- ☐ Child
Parent
- ☐ Child
- ☐ Parent
- ☐ Parent
Child

What operations can put a process into ready state?

1. A process is launched
2. A process is preempted by another
3. A process issued an I/O request and the I/O request is not done yet

- ☐ 1, 2
- ☐ 1
- ☐ 1, 2, 3
- ☐ 2, 3

If a process P creates several threads, which one is not shared by these threads

- ☐ Stack registers of threads
- ☐ Code section of process P
- ☐ Global variables of process P
- ☐ Open files of process P

Maximum marks: 3

2 Scheduling (2 Questions)

Starvation in scheduling means that some processes may not obtain resources perpetually for their execution. Select the one that consists of two starvation-free scheduling algorithms.

- ☐ FIFO, RR
- ☐ FIFO, SJF
- ☐ SJF, STCF
- ☐ SJF, RR

Task	Arrival time	Execution time
1	0	20
2	5	13
3	5	15
4	10	7

Assuming the task set consists of 4 tasks as above, which task would be completed second if STCF (Shortest Time to Completion First) scheduling algorithm is deployed?

- ☐ 1
- ☐ 4
- ☐ 3
- ☐ 2

Maximum marks: 2

3 Memory (2 Questions)

Which of the following statement is not true?

- ☐ Segments may lead to external fragmentation.
- ☐ Multiple segments need only one pair of base-bound registers.
- ☐ Code segment in address space is static and will not be changed at runtime.
- ☐ Arguments to a function is allocated to stack at runtime.

Given an architecture with 2KB address spaces and 16KB physical memory, the architecture performs dynamic relocation using base-bound registers. The base register has 0x00000150 (decimal: 344) and the bound/limit register has decimal 400. What is the physical address of virtual address 0x0400?

- ☐ 0x00000190
- ☐ 0x00000450
- ☐ Segmentation Fault/Violation
- ☐ 0x000002E4

Maximum marks: 2

4 Page (3 Questions)

Given a set of free memory partitions, 34, 26, 45, 36, which partition will be allocated to a process with memory requirement of 22 if using worst-fit algorithm? After the allocation, what will be the new free memory partitions?

- ☐ 45, (34, 26, 23, 36)
- ☐ 26, (34, 24, 45, 36)
- ☐ 36, (34, 26, 45, 14)
- ☐ 34, (12, 26, 45, 36)

Which of the following one is correct about page replacement policy?

- ☐ LRU always performs better than FIFO in terms of the number of page faults.
- ☐ For a given page access sequence, increasing the number of page frames always reduces page faults for all replacement policies.
- ☐ Multi-level page table is used to reduce the performance overhead.
- ☐ Upon first access of a page, two memory references will be made.

Which of the following is true?

1. If dirty bit of a page is set, it means the page has been modified and is different from the one on disk.
2. Demanding page loads required pages in advance.
3. Each process has its own page table.
4. Even using swapping, the virtual address size must be smaller than the physical address size.

- ☐ 1
- ☐ 1, 3
- ☐ 1, 2, 3
- ☐ 1, 2, 3, 4

5 Concurrency (4 Questions)

Given two processes, **P1** and **P2**, the two processes concurrently access a shared variable **x**. The following shows the instructions two processes use to update shared variable **x**

P1:

load R1, x // load x to register R1

add R1, R1, #1 // increment R1 by 1

store x, R1 // store R1 to x

P2:

load R2, x // load x to register R2

sub R2, R2, #1 // decrement R2 by 1

store x, R2 // store R2 to x

If **x** is initialized to **1** and both P1 and P2 manipulate **x** once, select the one that has **all possible** results. *Note that no presumed order for processes' execution.*

- ☐ 0, 1, and 2
- ☐ 1
- ☐ -1 and 3
- ☐ -1, 0, 1, and 2

What are the solution requirements for critical-section problems?

- ☐ Preemption, Progress, Bounded Waiting
- ☐ Mutual Exclusion, Progress, Bounded Waiting
- ☐ Mutual Exclusion, Preemption, Bounded Waiting
- ☐ Mutual Exclusion, Preemption, Progress

Which of the following statement is true regarding semaphores?

- ☐ `sem_post()` also blocks the caller
- ☐ Two semaphores are needed to implement a solution for producer-consumer problems.
- ☐ If a semaphore is initialized to 0, it is implemented as a binary lock
- ☐ Semaphores cannot have negative values

Which of the following is not a condition for deadlock?

- ☐ Hold-and-Wait
- ☐ Circular Queue
- ☐ Mutual Exclusion
- ☐ Non-Preemption

Maximum marks: 4

6 File Systems (2 Questions)

Which of the following is not the component of on-disk structures?

- ☐ Inode block
- ☐ Superblock
- ☐ Inode bitmap
- ☐ Sector

If a disk has 1 GB for data blocks and the block size is 4 KB, how many **blocks** does **data bitmap** need? **Note that the block size is still 4 KB**

- ☐ 8
- ☐ 16
- ☐ 64
- ☐ 32

Maximum marks: 2

7 Process (2 Questions)

```
int main() {  
    int i;  
    int ret = fork();  
    if(ret==0) {  
        printf("1");  
        exec(NEW_COMMAND); % NEW_COMMAND will not print anything  
    } else {  
        printf("2");  
    }  
    printf("3");  
}
```

Assume there is no error, i.e., the return value of fork will never be negative.

What is the output of executing the above code snippet? (1Point)

If we want to guarantee the parent executes until the completion of the child, where should a wait function be added? (1 Point)

(Type 1, 2, or 3 to indicate before printf("1"), printf("2"), and

printf("3"))

Maximum marks: 2

8 Scheduling (3 Questions)

Given a task set as follows,

Task	Arrival Time	Execution Time
A	0	17
B	3	21
C	6	32
D	9	15
E	11	22

If SJF is used to schedule the task set, what is the average waiting time? (1 Point)

The average waiting time is .

If RR is used to schedule the task set and **time quantum/time slice is 10**, what is the average turnaround time? (1 Point) (Hint: pay attention to adding task to the waiting queue.)

The average turnaround time is .

Again assuming using RR, if the time quantum is reduced to 5, will the average turnaround time be **increased** or **decreased**? (1 Point) (Input: increased or decreased in the answer box.)

The average turnaround time will be

Maximum marks: 3

9 Memory (2 Questions)

Two methods exist for allocating contiguous physical memory to a process.

What are the two methods? (1 Point)

Fill in your answer here

What are the drawbacks of these two methods? (1 Point)

Fill in your answer here

Maximum marks: 2

10 Page (4 Questions)

Given:

- Address space size: 16 KB
- Physical memory size: 64 KB
- Page size: 256 bytes

How many bits does the virtual address need for **page offset**? (1 Point)

Fill in your answer here

Assume virtual memory is managed with a **linear page table**. Given the following page table,

Page number (Decimal)	Frame number (hexadecimal)
0	0x00000051
1	0x0000006f
2	0x000000a5
3	0x0000003c
4	0x000000ea
5	0x000000c9
6	0x00000010
7	0x00000027
8	0x000000b4
9	0x00000064
10	0x0000001f
11	0x0000009a

What is the corresponding physical address of virtual address **0x0bbc**? (1 Point)

Fill in your answer here (The address should be in hexadecimal)

Assuming that the above page table entries have been loaded into TLB and TLB can store 20 entries. If process accesses the following virtual addresses, how many page faults does it have? (1 Point)

0x04db
0x07ad
0x03ef
0x06b8
0x0Ad7
0x0985
0x0D17
0x0B62
0x0264
0x007d
0x05fb

0x0876
0x012f
0x0C15

Fill in your answer here

If $P_{miss} = 1\text{ms}$ and $P_{hit} = 50\text{ns}$, what is the average memory access time for the above address sequences? (1 Point) (Only retain the first two decimal places when calculating miss rate)

Fill in your answer here

Maximum marks: 4

11 Concurrency (2 Questions)

Two threads access a shared variable x

Thread 1	Thread 2
<pre>int y; y = x * 2; // A x = y; // B</pre>	<pre>int z; z = x + 1; // C x = z; // D</pre>

Statements within a thread must execute sequentially, but statements from different threads can interleave in any order. If x is initially set to 3,

how many different values of x are possible after executing both threads once?

What are they? (2 Points)

Fill in your answer here

Thread 1	Thread 2
<pre>int y; sem_wait(S); y = x * 2; // A x = y; // B sem_post(S);</pre>	<pre>int z; sem_wait(S); z = x + 1; // C x = z; // D sem_post(S);</pre>

Again assume x is initially set to 3. If a binary semaphore S which is initialized to 1 is added as above,

how many values of x are possible after executing both threads once?

What are they? (2 Points)

Fill in your answer here

Maximum marks: 4

12 I/O Devices (3 Questions)

Two I/O communication methods were discussed in the lecture.

What are the two communication methods? (1 Point)

Fill in your answer here

How do they work, respectively? (1 Point)

Fill in your answer here

What is the advantages of one method over another? (You just need to write one.) (1 Point)

Fill in your answer here

Maximum marks: 0

13 File Systems (2 Questions)

Assume a computer has a hard disk drive as follows:

Metric	Performance
RPM	15000 rotations/min
Average Seek Time	5 ms
Max transfer	600MB/s

If you are reading a file of 4MB, answer the following questions:

What are the three key parts that contribute to the latency of reading the file? (1.5 Point)

Fill in your answer here

How long does reading this file take on average? (1.5 Point)

Pay attention to the time units

Fill in your answer here

Maximum marks: 3